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Intelligent analysis and recommendation of educational content in network engineering: a study on teaching effectiveness evaluation model based on knowledge reasoning and multimodal knowledge graph

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Abstract

With the rapid development of information technology, network engineering education is gradually moving towards intelligence, but existing teaching content analysis and recommendation systems still have limitations in personalized evaluation. Traditional methods mostly rely on rule-driven or single modal data, resulting in incomplete construction of the Knowledge Graph (KG), low accuracy of content recommendation, and insufficient coverage of knowledge points. In response to these issues, this article combined Knowledge Reasoning (KR) and Multimodal KG with Graph Neural Network (GNN) to construct a Knowledge Reasoning and Multimodal Knowledge Graph-based Graph Neural Network (KRMKG-GNN) intelligent teaching content analysis and recommendation model based on KR and Multimodal KG. This model constructs a multimodal KG that integrates text, image, and user behavior data through GNN, and then uses KR module to dynamically evaluate students' knowledge status and generate personalized recommendations. The experiment was validated on the learning data of 500 students, and the results showed that the model's recommendation accuracy reached 92.7%. Compared with Graph Attention Network (GAT), the knowledge point coverage has increased by 18.5% and the error rate has decreased by 1.6. Through deep integration and reasoning analysis of multidimensional data, this model effectively enhances the intelligence level of teaching effectiveness evaluation and has broad application potential.

Keywords Multimodal knowledge graph, Knowledge reasoning model, Personalized learning recommendation, Dynamic educational assessment, Graph neural network



1 Introduction

Driven by modern information technology, the field of education is gradually developing towards intelligence and personalization. In network engineering education, with the continuous enrichment of teaching content and learning resources, how to efficiently organize and recommend learning content suitable for students has become a research focus. Traditional teaching recommendation systems mostly rely on rule-based recommendation models or analysis of a single data source, which often exhibit significant limitations in complex teaching environments. Rule-based recommendation systems [1, 2] often require manual rule setting, which makes it difficult to cover diverse learning contexts, resulting in a lack of flexibility and adaptability in the recommended content. Data analysis based on a single mode [3, 4] cannot effectively use the multidimensional data of students in the learning process, and ignores the potential value of multimodal information such as video, text and behavioral data. This makes it difficult for existing systems to accurately capture the dynamic changes in students' knowledge levels when generating personalized recommendations [5, 6], and the evaluation of knowledge mastery tends to be static, making it difficult to support accurate tracking and feedback of long-term and complex learning processes. In terms of KR [7, 8], existing models often use logical reasoning or simple association rule mining techniques [9, 10]. These methods lack deep knowledge representation in the inference process, resulting in insufficient contextual adaptability of the inference results, making it difficult to cope with the changing needs in teaching scenarios. The evaluation of teaching effectiveness [11] also mostly relies on static data and linear statistical methods [12], lacking in-depth analysis of the dynamic changes in the learning process, resulting in evaluation results that often cannot reflect students' personalized needs and actual progress. Therefore, it is necessary to design an intelligent teaching content analysis and recommendation system that can integrate multimodal data [13] and has efficient KR capabilities [14, 15] to solve these problems.

In view of the fact that the current model relies on single-modal data and has poor dynamic adaptability, it is difficult to accurately capture the changes in students' knowledge level and realize personalized recommendations. This paper proposes a KR and multimodal KG teaching content intelligent analysis and recommendation model to fill the gaps in the existing system in data integration depth, reasoning dynamics and personalized recommendation capabilities. This model constructs a multimodal KG using GNN technology, which deeply integrates multidimensional data such as students' text, images, and behaviors to form a unified knowledge representation framework. On this basis, the KR module dynamically evaluates students' knowledge status, relies on deep learning algorithms to mine potential knowledge associations from complex multimodal data, and generates learning content recommendations based on students' personalized needs. To achieve this function, the model preprocesses and extracts features from data of different modalities, constructs KG reflecting students' knowledge structure through GNN, and finally uses an inference engine to dynamically analyze students' learning trajectory and knowledge mastery, generating personalized learning paths. The experiment shows that the model proposed in this article performs well in terms of knowledge coverage, recommendation accuracy, and evaluation dynamism, and exhibits strong adaptability and robustness in the long-term learning process. This article introduces multimodal KG and KR technologies in network engineering education, and through the combination of GNN and deep learning methods, achieves precise capture

and intelligent recommendation of students' dynamic needs, significantly improving the intelligence level and personalized recommendation ability of the teaching content recommendation system.

2 Related work

There are many methods available for educational content recommendation, and KR is one of them. It generates personalized learning paths by dynamically analyzing students' knowledge status, thereby improving the accuracy and adaptability of recommendations. By reviewing KR methods based on KG [16, 17], three main methods were analyzed: inference methods based on logical rules, representation learning [18], and neural networks [19, 20]. The results indicate that KR can effectively infer new knowledge and discover erroneous knowledge, improve the accuracy and practicality of KG, and provide support for applications such as vertical search and intelligent question answering [21]. This theoretical foundation provides support for practical applications. Li et al. used three different types of embedding models and their combination strategies: static embedding, contextual embedding of pre-trained language models, and contextual embedding of large language models, to conduct an empirical evaluation of educational resource recommendation, and achieved a model that performed well on both exercises, proving the feasibility of semantic text similarity based on text embedding in educational content recommendation [22]. On the other hand, Troussas et al. used the KG method to recommend appropriate educational content to learners by building an interconnected framework containing learners, educational resources and their relationships, and combined with contextual signals of learner characteristics, achieving highly effective content recommendation in an intelligent tutoring system for Java programming language learning [23]. These studies demonstrate the potential of KG and embedding models in the field of educational technology, and provide important theoretical basis and practical guidance for the optimization of personalized learning and intelligent recommendation systems [24].

For the research on teaching effectiveness evaluation [25, 26], some studies have adopted a results oriented approach and constructed a university teaching quality evaluation system around the quality of graduates [27], clarifying the two quantitative dimensions of academic level and employment competitiveness of graduates, establishing a development potential index evaluation model, and providing reference standards for self-evaluation of teaching effectiveness. At the same time, the composition of graduate quality and the construction method of evaluation system have been explored [28]. Subsequently, research explored the reform of ideological and political education evaluation in college English courses through a combination of online and offline teaching models [29–31], and proposed suggestions for improving the evaluation system to promote the improvement of educational quality [32]. Researchers adopted the teaching philosophy of "promoting learning through evaluation" and designed a management system including a teaching intelligence system [33], an evaluation and testing system, and an information feedback system. They combined interactive teaching methods to plan the interactive teaching process of college English courses. Finally, they collected four-dimensional data through the evaluation learning teaching mode and established a learning performance evaluation system based on big data. The experimental results

show that this teaching mode can significantly improve students' academic performance and practical English application ability [34].

Although these studies have made important contributions to the recommendation and teaching evaluation system of intelligent education [35], there are generally problems such as insufficient ability to integrate multimodal data and lack of dynamic KR capability. In response to these shortcomings, this article combines multimodal KG [36, 37] and KR [38], aiming to enhance the intelligence and personalization level of recommendation systems through more comprehensive data integration and deep reasoning.

3 Construction of teaching effectiveness evaluation model

3.1 Data preprocessing and feature extraction

This article collected multimodal learning data for the Network Engineering major from classroom teaching, online learning platforms, and laboratory practices at a certain school. There are text data, image data, and student behavior data. The text data comes from 500 students' classroom notes, homework submissions, and online quizzes, totaling approximately 1.5 million words. The image data comes from teaching video clips accessed by students during their online learning process, totaling 1200 video frames. Behavioral data records various behavioral trajectories of students during the learning process, including learning duration, resource access frequency, homework submission time, etc. A total of 5000 learning behavior records were obtained. Each type of data requires different methods for feature extraction during preprocessing.

When preprocessing text data, it is necessary to convert the text data into a digital vector representation. By using word embedding technology to represent words in text as vectors, assuming the dimension of word embedding is d , each text data is represented as a vector matrix with dimension d . If a single text data is T , the corresponding vector representation is:

$$X_T = [x_1, x_2, \dots, x_d], x_i \in \mathbb{R} \quad (1)$$

X_T represents the word vector matrix of the text, and x_i is the word vector of the i th word. The noisy words in text data need to be removed through stop word filtering and semantic analysis.

This article adopts image preprocessing and feature extraction techniques in the image data preprocessing process. The image can be grayscale processed, with a uniform image size of 224×224 pixels, and Gaussian filtering can be applied to remove noise. In the feature extraction process, Convolutional Neural Networks (CNN) [39] are used for feature extraction to extract key feature points for each image. Let the feature representation of image I be F_I , which is obtained through the operations of convolutional and pooling layers:

$$F_I = f_{pool}(f_{conv}(I, W)), F_I \in \mathbb{R}^k \quad (2)$$

f_{conv} represents convolution operation, W is convolution kernel parameter, f_{pool} is pooling operation, and k is dimension.

In behavioral data preprocessing, standardization methods are used to normalize behavioral data from different dimensions, so that the model can handle different data scales. This paper standardizes behavioral data such as learning duration, resource access frequency, and homework submission time to reduce model processing errors

caused by different data scales. On this basis, in order to further improve the dynamic analysis capabilities of behavioral data, the Dynamic Time Warping (DTW) algorithm is introduced to accurately capture changes in their behavioral patterns by measuring the similarity of learning behavior trajectories between students. The DTW method measures the differences between two students in behaviors such as learning duration and resource access frequency, and calculates their minimum difference in the time series. Assuming that the behavior data is recorded as B , including learning duration t , resource access frequency f , and job submission time s , the behavior data matrix can be represented as:

$$X_B = [\frac{t - \mu_t}{\sigma_t}, \frac{f - \mu_f}{\sigma_f}, \frac{s - \mu_s}{\sigma_s}] \quad (3)$$

μ and σ respectively represent the mean and standard deviation of behavioral data, and X_B is the standardized behavioral data vector.

The final step of feature extraction is to uniformly map all data to multimodal KG [40]. After the feature extraction of text, image, and behavioral data is completed, each type of data is represented by feature vectors and enters the KG node. By using GNN [41, 42], these feature vectors were further integrated into a unified feature representation [43]. The feature extraction formula is as follows:

$$h_i^{(l+1)} = \sigma(\sum_{j \in \mathcal{N}(i)} \frac{1}{c_{ij}} W^{(l)} h_j^{(l)}) \quad (4)$$

$h_i^{(l)}$ is the feature representation of the i th node in the l th layer, and $\mathcal{N}(i)$ is the set of neighboring nodes connected to the i th node. c_{ij} is the normalization constant between nodes i and j , and $W^{(l)}$ is the weight matrix of the l th layer. The σ here is the activation function. The data distribution results generated during the preprocessing and feature extraction processes are shown in Table 1.

3.2 Multimodal KG construction

Multimodal KG is a knowledge representation structure that integrates multiple modal data (text, image, behavioral data), and uses methods such as GNN to transform data from different sources into unified nodes and edges for subsequent KR and recommendation tasks. The key steps in building a multimodal KG include node definition, edge construction, and fusion of multimodal data. The schematic diagram of the structure of multimodal KG is shown in Fig. 1.

The blue nodes shown in Fig. 1 are text nodes, the green nodes are image nodes, and the red nodes are behavioral data nodes. The edges connecting the nodes represent the correlation between them, and the weights of the edges represent the strength of the correlation.

Table 1 Distribution of data preprocessing and feature extraction results

Data type	Data volume	Feature dimension	Noise removal ratio	Processed data volume
Text data	1.5 million words	300 dimensions	5%	1.425 million words
Image data	1200 frames	512 dimensions	3%	1164 frames
Behavior data	5000 records	3 dimensions	None	5000 records

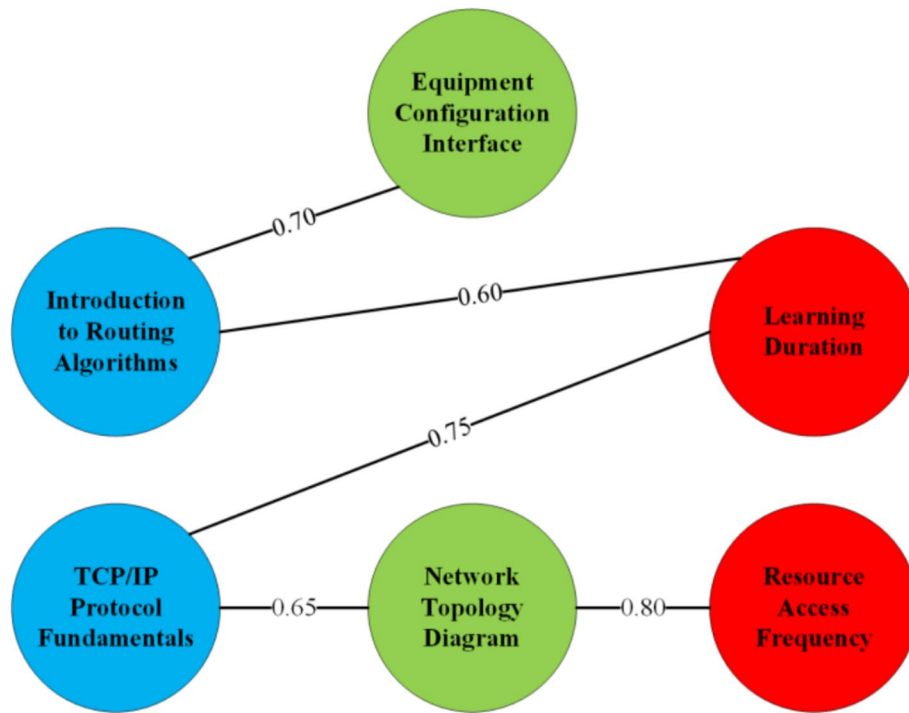


Fig. 1 Schematic diagram of the structure of multimodal KG

3.2.1 Definition and construction of nodes

The nodes in KG represent students' knowledge points, and the edges are constructed through students' learning paths in different modal data. In this study, knowledge points represent specific topics or concepts rather than individual lessons. Each node contains the student's learning performance on that knowledge point, which is used to reflect the student's understanding of the topic. In the knowledge graph, nodes represent knowledge points rather than specific text, images, or behavioral data. The latter is only integrated into the node as feature information to help analyze the relationship between learning behavior and knowledge mastery.

Assuming that the text, image, and behavioral data of a certain knowledge point are T_i , I_i , and B_i , respectively, they form feature vectors after feature extraction. The characteristic representation h_i of the node is:

$$h_i = [T_i, I_i, B_i] \quad (5)$$

When constructing knowledge graph nodes, the feature vectors of each data type (text, image, and behavior data) have different dimensions and semantic information. In order to enable GNN to uniformly process these feature vectors, this study adopts a multi-layer feature conversion strategy to map the features of different modalities to a unified embedding space. In text data, word vectors or pre-trained language models are used to embed text into a vector representation of fixed dimensions; for image data, CNN is used to extract image features and then linearly transforms them to the same dimension as the text features; behavior data is normalized by standardization. Finally, the features of each modality are combined to form a node embedding into the GNN through weighted summation, so as to realize the update and propagation of node representation in the graph structure. This method not only retains multimodal information in the

graph structure, but also allows the model to flexibly update and optimize the relationship between different modalities during reasoning.

3.2.2 Construction of edges

The edges between nodes reflect the correlation between different knowledge points. The weight of edges is determined by the transition probability of students between knowledge points and the semantic similarity between knowledge points [44]. GNN can be used to optimize the weights of these edges to construct a more accurate KG structure. The weight between nodes i and j is w_{ij} , and its calculation formula is:

$$w_{ij} = \frac{\text{sim}(h_i, h_j)}{\sum_{k \in N(i)} \text{sim}(h_i, h_k)} \quad (6)$$

$\text{sim}(h_i, h_j)$ represents the feature similarity between nodes i and j , while $N(i)$ represents the set of neighboring nodes connected to node i .

When constructing multimodal KG, the correlation between nodes is determined by semantic similarity and the transition probability of learning paths. Table 2 shows the strength and weight of the correlation between some knowledge points.

3.2.3 Multimodal data fusion

The fusion process of multimodal KG is described by the following formula:

$$h_i^{(l+1)} = \sigma\left(\sum_{j \in N(i)} w_{ij} W^{(l)} h_j^{(l)}\right) \quad (7)$$

3.2.4 Multimodal KG construction results

The construction of multimodal KG relies on deep integration of different types of data and reasonable construction of node relationships. After preprocessing and feature extraction, text, image, and behavioral data form the basic nodes of KG. The feature information of each type of node is different, but they together constitute a comprehensive representation of students' knowledge status. In the process of constructing nodes in KG, the weights of edges reflect the interrelationships between various knowledge points. The higher the weight, the stronger the correlation between knowledge points. Table 3 shows the number of different modal data nodes, feature dimensions, and connections between nodes in the constructed multimodal KG.

The data in Table 3 shows that the number of text data nodes is the highest, reaching 1500 with an average feature dimension of 300, while the number of image data nodes is second, with 1200, and the average feature dimension is significantly higher, reaching 512. This indicates that the information content of image data in KG is relatively rich. The behavior data has the least number of nodes, but the average weight of its edges is the highest, at 0.75, indicating that although the dimension of behavior data is small, it

Table 2 Correlation strength of knowledge points

Knowledge point A	Knowledge point B	Similarity	Weight
TCP protocol	Network security	0.85	0.75
Packet transfer	Routing protocol	0.90	0.80
Network topology	Routing protocol	0.78	0.70
Network security	Firewall technology	0.88	0.77
Data encryption	Firewall technology	0.82	0.73

Table 3 Results of multi modal KG construction

Node type	Number of nodes	Average feature dimension	Number of edges	Average edge weight
Text data nodes	1500	300	1200	0.65
Image data nodes	1200	512	1000	0.7
Behavior data nodes	500	3	800	0.75
Total	3200	–	3000	–

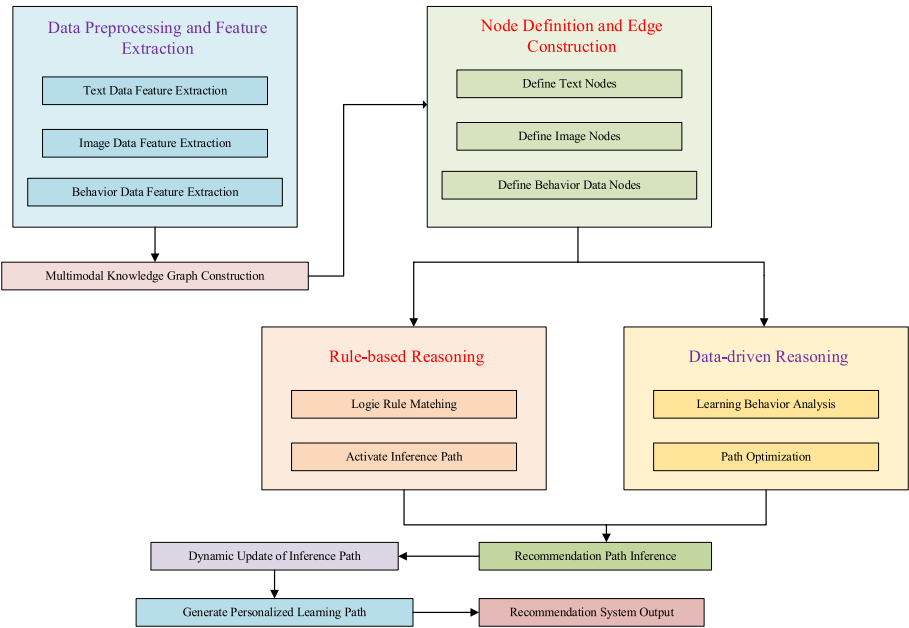


Fig. 2 KR model design flow chart

has strong explanatory power for the correlation between knowledge points. A total of 3200 nodes and 3000 edges indicate that multimodal KG has a comprehensive knowledge representation structure, effectively reflecting the complex relationships between different modal information.

3.3 KR model design

To improve the accuracy of personalized recommendations, this article combines rule-based reasoning and data-driven reasoning to design an inference engine. The inference engine based on multimodal KG dynamically analyzes students’ knowledge status and generates personalized learning paths [45, 46]. The KR model design process is shown in Fig. 2.

3.3.1 Rule reasoning

Rule reasoning is mainly based on preset logical rules, which match students’ knowledge status with the association relationship in KG to determine the recommended path of learning content that meets the conditions. If one node in KG represents knowledge point K_i and another node K_j represents its related knowledge points, then in a certain learning state, if the conditional rule R is satisfied, the correlation between nodes K_i and

K_j can be activated, forming an inference path. The mathematical expression for the process of rule-based reasoning is:

$$P_{ij} = f(K_i, K_j, R) \quad (8)$$

P_{ij} is the weight of the inference path between nodes K_i and K_j , and f is the application function of the inference engine to rule R .

A short example is shown below: Consider a network engineering course where one of the key knowledge points is the “TCP protocol”. In our model, this knowledge point is represented as a node in a multimodal Knowledge Graph (KG). This node encapsulates various forms of data related to the TCP protocol, including textual descriptions in handouts, images in educational materials, and behavioral data indicating student interactions with relevant learning resources. These multimodal data points are integrated into the KG's TCP protocol node to create a rich representation of knowledge points that includes content-related features and student interaction data.

The KR module dynamically evaluates student interactions with TCP protocol nodes. It uses rule-based reasoning to identify patterns that indicate mastery or comprehension gaps. For example, if a student frequently accesses detailed diagrams of TCP, but rarely uses troubleshooting material, the model may infer the need for more applied learning experiences. Based on the combined data and KR results, student performance on the TCP protocol is categorized into predefined categories such as “mastery”, “partial mastery”, or “needs improvement”. This categorization is based not only on test scores, but also on a holistic view of students' engagement and interaction with various aspects of the TCP. For example, a student who is highly engaged with visual material but performs low on a quiz may be categorized as “partially mastered,” indicating the need for targeted interventions to strengthen specific areas.

Classification results directly impact the generation of personalized learning paths. Students categorized as “Needs Improvement” on the TCP protocol may receive suggestions for additional resources, such as interactive simulations or targeted practice, to enhance their understanding.

3.3.2 Data-driven reasoning and path calculation

Data-driven reasoning is based on students' actual learning data, predicting the next step of learning content through multimodal information such as behavioral data, learning progress, and test results.

$$S(K_i, K_j) = \alpha \cdot \frac{T_{ij}}{T_{total}} + \beta \cdot \frac{F_{ij}}{F_{total}} + \gamma \cdot \frac{P_{ij}}{P_{total}} \quad (9)$$

T_{ij} represents the learning duration of students between knowledge points K_i and K_j , F_{ij} represents access frequency, P_{ij} represents test scores, α , β , and γ are weight coefficients used to adjust the impact of various behavioral data on path selection.

3.3.3 Dynamic update of inference path

The inference engine optimizes the inference path through an adaptive [47] algorithm to ensure that the recommendation results after each inference [48] conform to the current knowledge state of the students. The formula for updating path weights is:

$$W_{ij}^{new} = W_{ij} + \eta \cdot \nabla_W L(W_{ij}) \quad (10)$$

$L(W_{ij})$ is the inference loss function, and η is the learning rate. By iteratively adjusting the path weights, the inference engine gradually optimizes the recommendation results.

3.4 Teaching effect evaluation model

When dealing with uncertainty in model prediction, this paper uses Bayesian methods or ensemble learning to quantify prediction risks, thereby improving the robustness of the model. Multi-model ensemble prediction can reduce the bias caused by a single model and generate more stable prediction results. For the purpose of dealing with potential bias in the data set, fairness constraints should be introduced in the data collection stage, and combined with debiasing algorithms or balanced data sampling methods to correct the problem of uneven sample distribution, thereby effectively improving the generalization ability of the model under different data distributions.

In order to accurately capture students' learning behavior, mastery of knowledge points, and personalized learning paths in teaching effectiveness evaluation, this article uses Bayesian Knowledge Tracing (BKT), multi-objective optimization, and reinforcement learning methods to construct a dynamic evaluation model. This model dynamically tracks students' learning progress, optimizes paths based on cognitive load and coherence between knowledge points, and continuously adjusts recommendation strategies through reinforcement learning to achieve personalized learning content recommendations. This model can accurately evaluate students' learning status and maximize learning benefits through optimized learning paths, thereby improving overall teaching effectiveness.

This paper uses DTW to measure the similarity of learning behaviors between different students. Assuming that the learning behavior trajectories of two students, A and B , are $X_A = \{x_{A1}, x_{A2}, \dots, x_{An}\}$ and $X_B = \{x_{B1}, x_{B2}, \dots, x_{Bm}\}$ respectively, the similarity measurement formula is:

$$DTW(X_A, X_B) = \min(\sum_{i=1}^n \sum_{j=1}^m d(x_{Ai}, x_{Bj})) \quad (11)$$

$d(x_{Ai}, x_{Bj})$ represents the difference between student A and B in the i and j behaviors, while DTW measures the minimum difference in their behavior time series. The system uses this method to identify the similarity of students' behavioral patterns during the learning process, and then infer their learning effectiveness.

The evaluation of students' mastery of knowledge points is not limited to test scores and homework completion, but also needs to consider feedback data during the learning process. This article introduces the BKT model to dynamically track students' performance during multiple learning processes and predict their mastery of a certain knowledge point. The core formula of the BKT model is as follows:

$$P(L_{t+1}|L_t, O_t) = \frac{P(O_t|L_t)P(L_t)}{P(O_t)} \quad (12)$$

$P(L_{t+1})$ represents the probability of students mastering the knowledge points after the $t + 1$ th learning session, $P(L_t)$ represents the probability of mastery during the t th learning session, $P(O_t|L_t)$ represents the conditional probability of students' observed

performance during the t th learning session, and $P(O_t)$ represents the marginal probability. The system continuously adjusts students' assessment of their mastery of knowledge points through Bayesian update formulas, thereby reflecting their learning progress in real time.

To recommend personalized learning paths, it is necessary to consider the current learning status of students, as well as the cognitive load and complexity of knowledge points during the learning process. Therefore, this article introduces a multi-objective optimization strategy to balance learning effectiveness and cognitive load.

Assuming that the set of knowledge points that students need to learn is $K = \{K_1, K_2, \dots, K_n\}$, the optimization objectives of learning path $P(K_i, K_j)$ include minimizing students' cognitive load $C(P)$ and maximizing the coherence $Q(P)$ between knowledge points. The optimization objective function is shown in formulas (13) to (14):

$$\min(C(P)) = \sum_{i=1}^n w_i C(K_i) \quad (13)$$

$$\max(Q(P)) = \sum_{i=1}^{n-1} \text{sim}(K_i, K_{i+1}) \quad (14)$$

w_i is the corresponding weight, and through multi-objective optimization algorithms, the model can recommend the most suitable learning path for students, ensuring that their learning process is moderately loaded and there is strong logical correlation between knowledge points.

The system combines reinforcement learning methods to dynamically adjust recommendation strategies by collecting performance data of students on different learning paths, improving the accuracy of teaching effectiveness evaluation. The goal of reinforcement learning is to maximize students' long-term learning benefits, which are defined as:

$$R = \sum_{t=0}^T \gamma^t r_t \quad (15)$$

r_t is the immediate feedback from students in the t th learning session, γ is the discount factor, and T is the total time step of learning. The system continuously updates students' feedback data on various knowledge points, optimizes recommendation strategies to improve students' learning efficiency and knowledge mastery.

3.5 Personalized recommendation algorithm design

The design of personalized recommendation algorithm is based on KR and multimodal KG, aiming to provide learning content recommendations that match the specific needs of students. Multimodal KG integrates different types of data such as text, images, and user behavior to form a complex knowledge network. Based on this, the KR module analyzes students' current knowledge status to dynamically adjust personalized recommendation content. The system infers recommended paths based on the performance information of students at each node in KG, as well as the correlation weights between nodes.

The core of recommendation algorithm is the rating function, and the rating function $R(u, i)$ is used to calculate the user u 's preference value for learning content i . Its expression is:

$$R(u, i) = w_1 \cdot f_{text}(u, i) + w_2 \cdot f_{image}(u, i) + w_3 \cdot f_{behavior}(u, i) \quad (16)$$

$f_{text}(u, i)$, $f_{image}(u, i)$, and $f_{behavior}(u, i)$ represent students' performance in text, image, and behavioral data, respectively, while w_1 , w_2 , and w_3 are the weight parameters for each modality. These weights are dynamically adjusted based on users' learning preferences in different modalities.

The calculation of recommendation weight is based on the semantic similarity between nodes and the performance in the learning path. Assuming that the similarity between two knowledge points K_i and K_j in KG is $sim(K_i, K_j)$, the formula for calculating recommendation weight W_{ij} is as follows:

$$W_{ij} = \frac{sim(K_i, K_j)}{\sum_{k \in N(i)} sim(K_i, K_k)} \quad (17)$$

$N(i)$ is a set of nodes connected to K_i , and during the recommendation process, priority is given to selecting paths with higher weights to generate a list of recommended learning content.

4 Teaching effect evaluation experiment

4.1 Experimental environment

This experiment was conducted on a high-performance computing platform to ensure the ability to handle complex model training and large-scale dataset analysis. Table 4 lists the hardware and software configurations used in the experiment, covering key elements such as processor, graphics card, memory, storage devices, operating system, and programming environment.

4.2 Experimental evaluation indicators

In order to comprehensively evaluate the effectiveness of the teaching content analysis and recommendation model in this article, accuracy (Acc), inference accuracy (IA), knowledge coverage (COV), inference efficiency (Efficiency), and error rate (Err) were used as evaluation indicators in the experimental evaluation process. These indicators are evaluated from the perspectives of model prediction accuracy, personalized

Table 4 Experimental environment configuration table

Component	Description	Specification	Remarks
CPU	Central Processing Unit (CPU)	Intel Core i9-11900 K, 3.5 GHz, 8 cores	Multi-threaded processing for model computations
GPU	Graphics Processing Unit (GPU)	NVIDIA RTX 3090, 24 GB GDDR6X	For parallel processing and deep learning tasks
Memory	Random Access Memory (RAM)	64 GB DDR4	To handle large-scale data and prevent bottlenecks
Storage	Solid-State Drive (SSD)	2 TB NVMe SSD	High-speed storage for fast data access and storage
Operating System	Software platform	Windows 10	Optimization Environment for Machine Learning Based on Windows
Programming Environment	Programming Languages & Libraries	Python 3.8, TensorFlow 2.4, PyTorch 1.8	For model development, training, and evaluation

recommendation quality, knowledge point coverage, and learning path optimization effectiveness.

Acc is used to evaluate the overall correctness of the model's prediction of students' knowledge status and measure the predictive performance of the model after multi-modal data fusion. The formula is:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \quad (18)$$

TP and TN represent the number of correctly predicted knowledge points that have been mastered and not mastered, while FP and FN represent the number of incorrectly predicted knowledge points that have been mastered but not actually mastered, and the number of incorrectly predicted knowledge points that have been mastered but not actually mastered.

IA represents the inference accuracy of the system in top-N results, with the following formula:

$$IA(N) = \frac{T_c}{T_{total}} \times 100 \quad (19)$$

T_c represents the number of correct reasoning results, and T_{total} represents the total number of reasoning iterations.

Cov is used in this article to measure the range of knowledge points covered by the system, and the formula is as follows:

$$Cov = \frac{K_{covered}}{K_{total}} \times 100\% \quad (20)$$

$K_{covered}$ is the number of knowledge points covered by the system, and K_{total} is the total number of knowledge points in the knowledge base.

Eff is evaluated by calculating the number of inferences completed per unit time, using the following formula:

$$Eff = \frac{I_{total}}{T_{processing}} \quad (21)$$

I_{total} is the total number of inferences, and $T_{processing}$ is the total processing time of the inference process.

Err measures the proportion of erroneous results in an inference system, and the formula is as follows:

$$Err = \frac{T_{error}}{T_{total}} \quad (22)$$

T_{error} is the number of erroneous inference results, and T_{total} is the total number of inferences.

5 Result analysis

5.1 Model recommendation performance analysis

In this experiment, this article tracked and recorded the Acc during the model training process to evaluate the performance of the model at different iterations. Figure 3 visually illustrates the trend of Acc variation in the model.

In Fig. 3, the vertical axis represents the prediction accuracy of the model, and the horizontal axis represents the number of training iterations. The accuracy of the model rapidly increases from around 73% to 85% in the first 20 iterations. Subsequently, the accuracy growth tended to flatten out and fluctuated around 92.7%, with small fluctuations that clearly reflected the stability of the model in the later stages of training, indicating that the model had basically converged.

5.2 Knowledge point coverage improvement

This article compares and analyzes the performance of GNN, Graph Convolutional Network (GCN), GAT, and the proposed KRMKG-GNN in terms of knowledge point coverage. The results are shown in Fig. 4. The vertical axis of this figure represents the percentage of knowledge point coverage. The higher the coverage, the more comprehensive the learning content recommended by the model. The horizontal axis shows the names of different models.

Figure 4 shows that the knowledge point coverage of the GNN model is 65.4%, the coverage of the GCN model is 70.1%, and the coverage of the GAT model is 75.9%. However, the KRMKG-GNN model proposed in this article achieves 94.4%, which is 18.5% higher than the GAT model. These data indicate that the KRMKG-GNN model effectively integrates multimodal data with intelligent reasoning, enabling the system to more

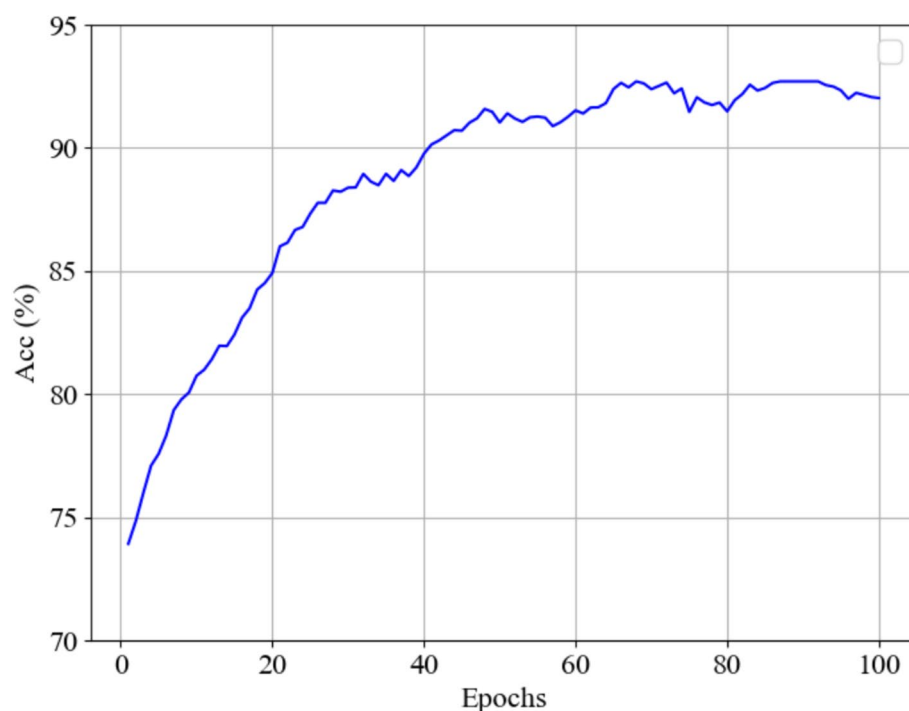


Fig. 3 Performance of the model during training iterations

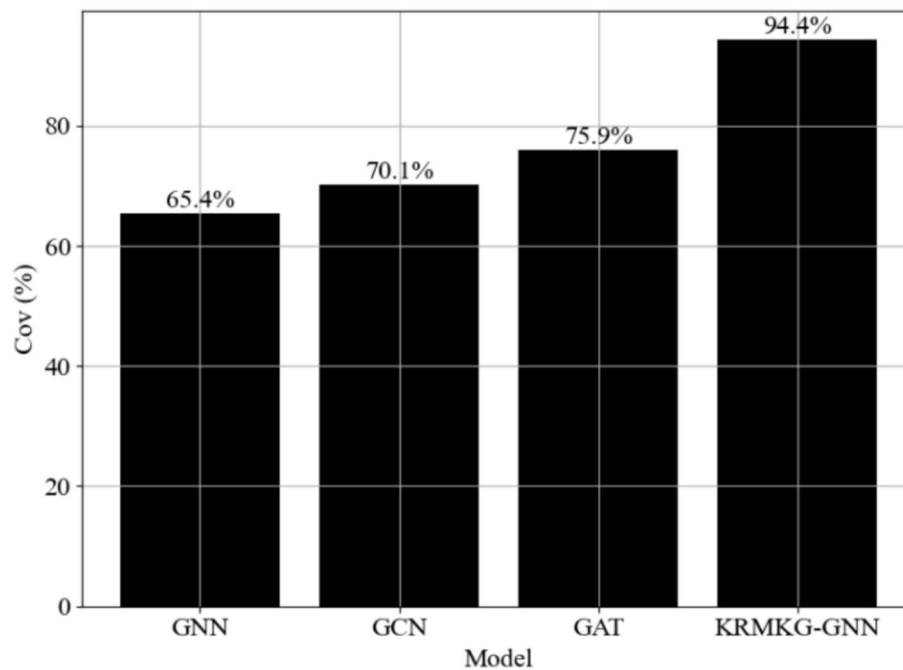


Fig. 4 Comparison of coverage rates of different models

Table 5 Algorithm performance on personalized learning path

Algorithm	IA (%)	Cov (%)	Eff	Err
GNN	82.5	65.4	0.65	4.5
GCN	85.7	70.1	0.69	3.8
GAT	89.1	75.9	0.72	3.1
GCLN	93	89.5	0.81	2
LLM-GNN	94.5	95.2	0.87	1.6
KRMKG-GNN	95.3	94.4	0.85	1.5

comprehensively and accurately cover students' knowledge points, thereby significantly improving the coverage of knowledge points.

5.3 Performance comparison of different algorithms

When evaluating personalized recommendation systems, different algorithms show obvious differences in knowledge coverage and reasoning accuracy. In order to have a more comprehensive understanding of the effectiveness of these algorithms, this paper experimentally compares the above four recommendation algorithms with the self-supervised learning-based Graph Contrastive Learning Network (GCLN) and the Large Language Model-enhanced Graph Neural Network (LLM-GNN). The results are shown in Table 5.

The KRMKG-GNN model proposed in this paper performs well in various indicators, with IA, Cov, and Eff of 95.3%, 94.4%, and 0.85 respectively. Compared with LLM-GNN, KRMKG-GNN's Err is reduced by 0.1 and IA is reduced by 0.8%, but Cov and Eff are slightly lower than LLM-GNN. These data show that KRMKG-GNN has shown strong comprehensive strength and maintained a high level in many indicators.

5.4 Response time analysis of recommendation system

In order to study the response performance of recommendation systems under different data scales, this article recorded the response time of the system on 1000, 5000, 10,000, 15,000, 20,000, and 25,000 records, as shown in Fig. 5. The horizontal axis represents the amount of data, and the vertical axis represents the response time of the system.

Figure 5 shows a positive correlation between system response time and data size. When there are 1000 pieces of data, the system response time is 375 ms. As the amount of data increases, the response time also increases. With 25,000 pieces of data, the response time reaches 537 ms. This indicates that the amount of data affects the computational efficiency of the system, suggesting that there is still room for improvement in the system's processing performance and scalability. To improve the scalability of the model, a distributed computing architecture and parallel processing technology should be used to share the data processing pressure and reduce the response time. Use high-performance graphics processors or tensor processors to accelerate large-scale data analysis, improve computing efficiency, optimize algorithms to reduce complexity, improve memory utilization, and reduce network communication overhead, so that the system remains efficient in a large-scale data environment.

5.5 Learning behavior data

Teachers analyze students' learning behavior data to better understand their learning habits, thereby optimizing teaching content and personalized learning plans. By analyzing key indicators such as learning time and access frequency, differences in students' learning patterns are revealed, providing data support for improving teaching methods. Figure 6 shows the feature distribution of these learning behaviors.

Figure 6 is a scatter plot of student learning behavior data, where the horizontal axis represents learning time and the vertical axis represents access frequency. Each point represents a student's learning behavior characteristics. The main point clusters are concentrated in the range of 2 to 6 h of learning time and 20 to 40 visits, indicating that most students perform similarly within this range. Some students study for more than 8 h but have a lower frequency of visits, while others have a shorter study time but a higher frequency of visits. As the learning time increases, some point groups show a

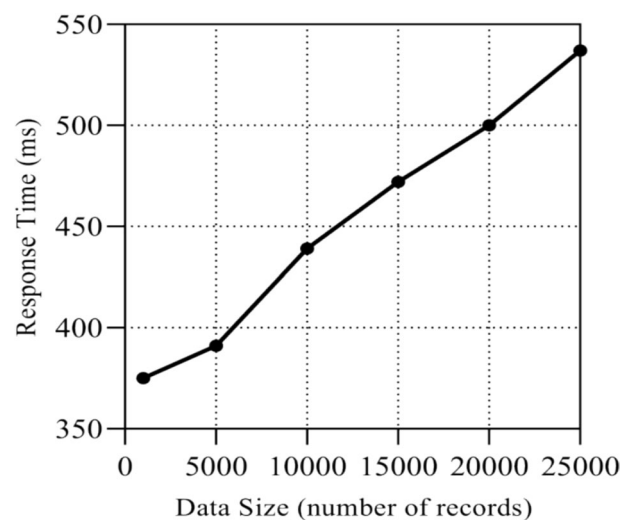


Fig. 5 System response time

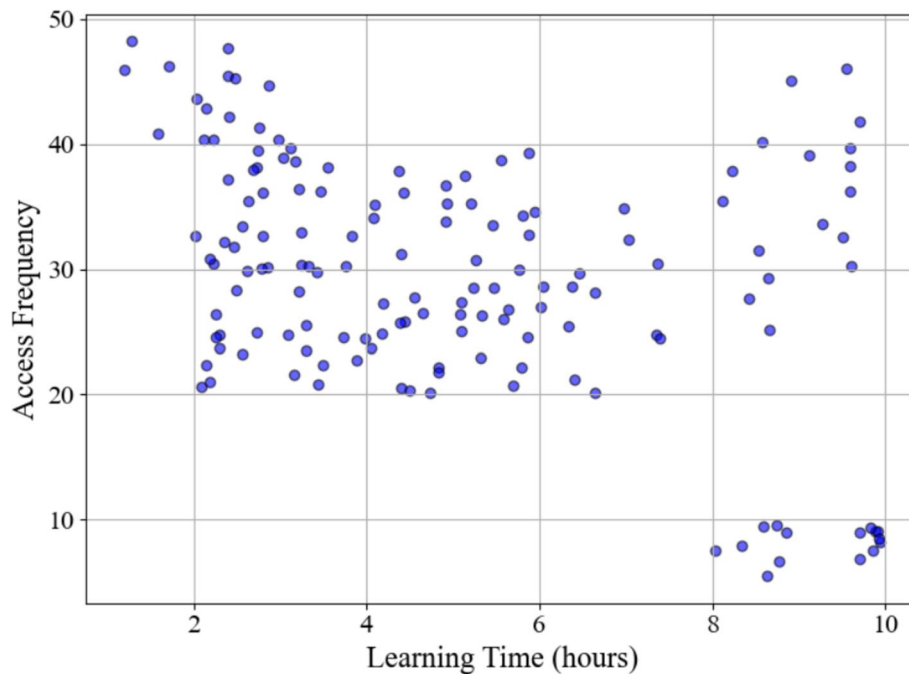


Fig. 6 Scatter plot of feature distribution of learning behavior data

trend of gradually increasing access frequency, indicating that students who invest more time in learning often have an increase in the number of times they access resources. These data provide valuable basis for understanding the learning patterns of different students.

5.6 Optimization effect of learning paths

The traditional learning path adopts a rule-driven approach, which is difficult to respond to students' personalized needs in real time based on preset learning content and fixed path arrangements. The learning path optimization scheme based on data analysis and dynamic feedback in this article dynamically adjusts the learning path by continuously tracking the learning progress and combining it with artificial intelligence algorithms [49, 50]. To demonstrate the improvement after path optimization, this article evaluated the learning effectiveness before and after optimization, as shown in Fig. 7. The horizontal axis represents the number of weeks of study, and the vertical axis represents the learning performance.

Figure 7 shows that the learning performance score before optimization increased from 60 points in week 1 to 70 points in week 10, while the optimized score rapidly increased from 60 to 76 points in week 5 within the same time range, ultimately reaching 90 points in week 10. This indicates that the optimized path more effectively promotes learning outcomes, especially showing significant improvement in the early stages.

5.7 Teaching effectiveness of the model in this article

In order to verify the effectiveness of the model in teaching effectiveness evaluation, this article conducted a semester of different teaching methods on students from five classes (with consistent overall levels), and compared the average final exam scores and classroom participation of each class. The results are shown in Fig. 8. The vertical axes of the

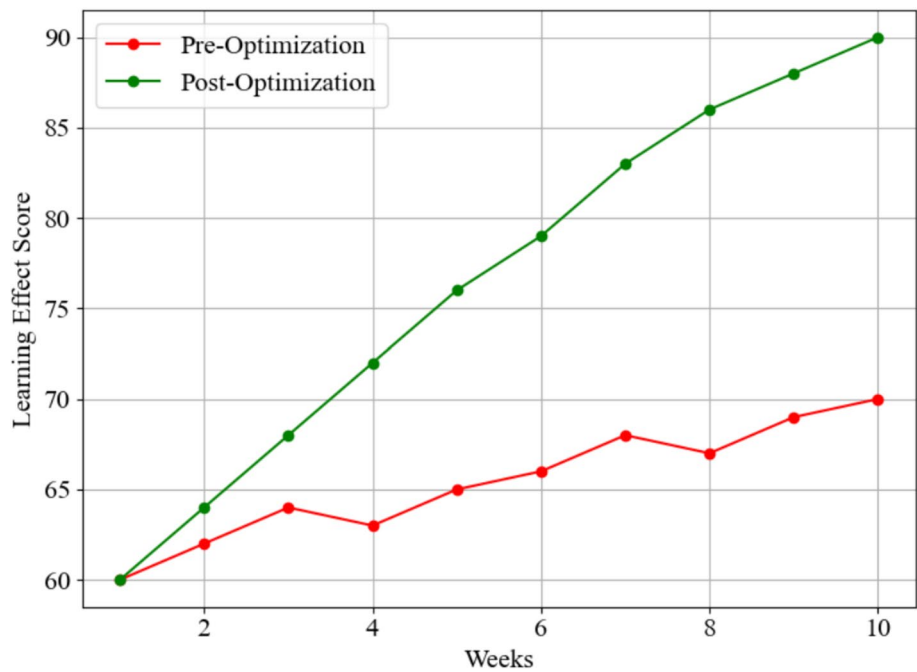


Fig. 7 Line chart comparing before and after learning path optimization

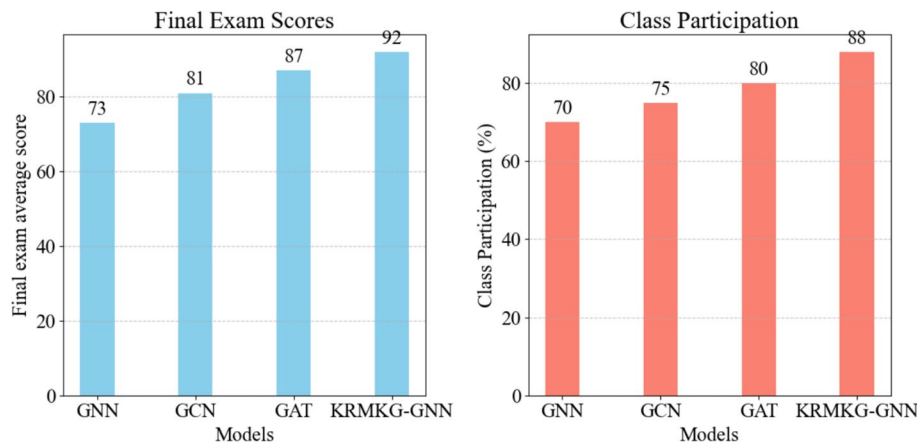


Fig. 8 Comparison of teaching effects

left and right sub-graphs are the students' average final exam scores and class participation, respectively, and the horizontal axes represent different teaching models.

Figure 8 shows the average final exam scores and classroom engagement for four different models. The class that adopted the KRMKG-GNN model for teaching achieved the highest average score of 92 in the final exam, higher than GAT's score of 87. In terms of classroom participation, the KRMKG-GNN model also leads with 88% participation, while GNN and GCN have participation rates of 70% and 75%, respectively. These data indicate that KRMKG-GNN is significantly effective in improving student engagement and exam performance.

5.8 Optimization effect of dynamic reasoning path

During the dynamic update process of the inference path, as the number of updates increases, the path weights gradually optimize. By tracking 50 updates, it is possible to visually observe the trend of changes in the inference engine at each stage. The variation of path weights is not only related to the training effectiveness of the model, but also affected by random noise. Figure 9 shows the weight change trend of the inference path in dynamic update. The vertical axis represents the value of the path weight, and the horizontal axis represents the number of updates.

As shown in Fig. 9, the initial path weight is 0.53, and as the number of updates increases, the weight gradually increases. In the 10th update, the weight reached about 0.65, while in the 30th update, the weight increased to around 0.77. The overall weight showed a gradually stable trend, reaching approximately 0.82 in the 50th update, indicating that the inference path gradually stabilized after optimization. The noise is small and the weight changes smoothly, reflecting the gradual improvement of recommendation accuracy.

5.9 Benefit analysis of reinforcement learning strategies

The cumulative learning benefit is the sum of rewards accumulated by the reinforcement learning model at each time step, used to measure the performance of the model in long-term learning. This article investigates the changes in benefits of reinforcement learning strategies at different time steps, as shown in Fig. 10.

In Fig. 10, the cumulative learning income shows an overall upward trend with the increase of time steps. In the first 20 time steps, the income growth is more significant, gradually increasing from about 15 to nearly 60. As the time steps increase, the growth rate of income slows down, reaching about 92 at the 100th time step. This growth trend indicates that the model's revenue increases rapidly in the initial learning stage and

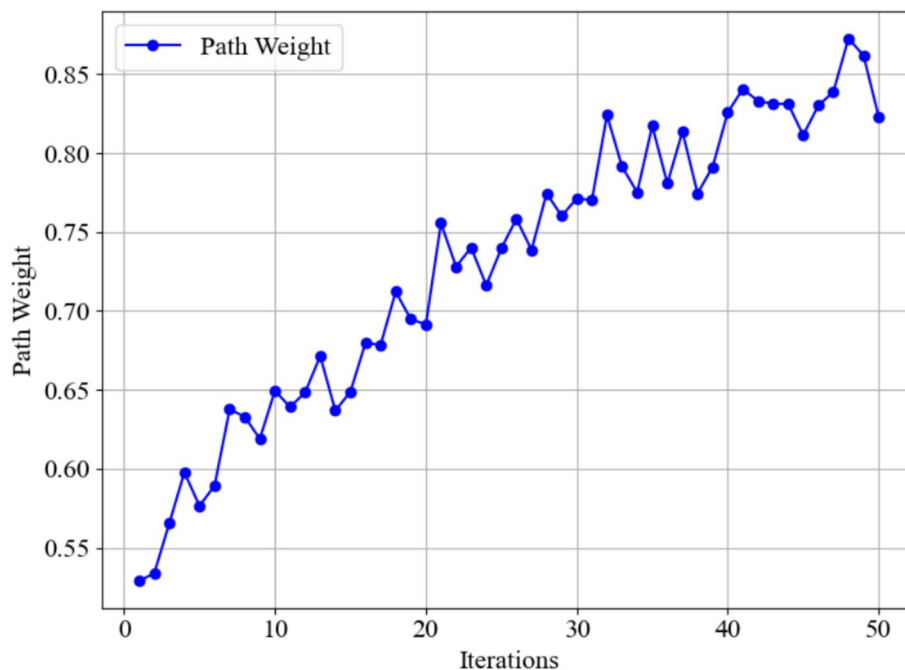


Fig. 9 Dynamic update process of inference path

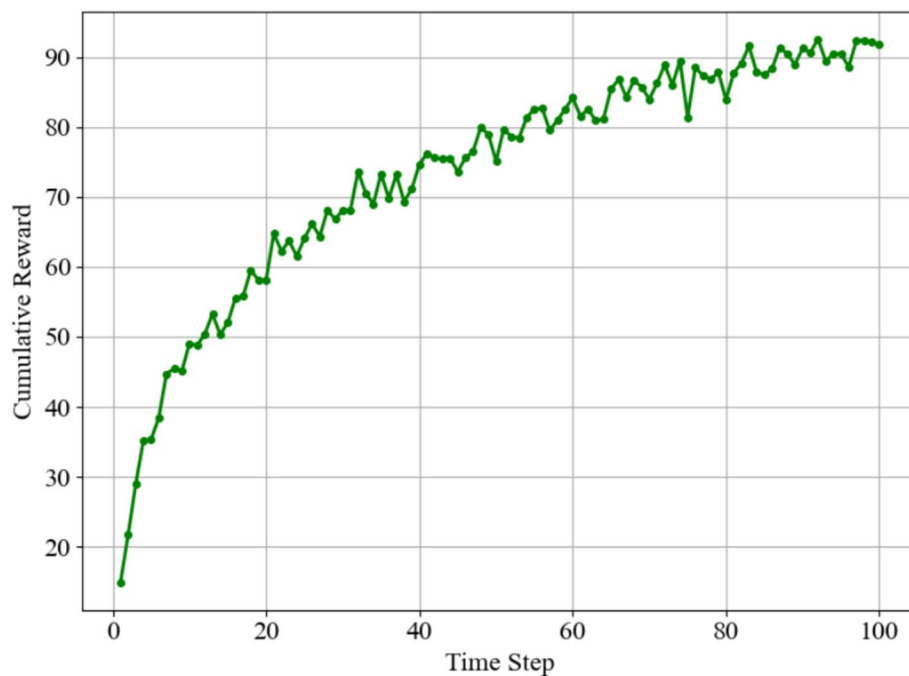


Fig. 10 Growth trend of learning revenue

gradually stabilizes thereafter. The results indicate that the influence of noise is relatively small, and the learning process of the model exhibits good stability.

6 Conclusions

The KRMKG-GNN model constructs a comprehensive multimodal knowledge graph by integrating text, image and user behavior data. This deep integration of data provides a richer representation of students' learning behaviors and knowledge states, which improves the accuracy and personalization of the model's recommendations for learning content. The KR module is able to dynamically assess students' knowledge states and generate personalized learning paths based on them. This dynamic reasoning capability enables the model to respond to students' learning progress and changing needs in real time, providing recommendations that are more closely aligned with individual learning trajectories. Meanwhile, Multimodal KG not only contains structured knowledge points, but also enhances the association between knowledge points through unstructured data (e.g., images and behavioral data), enabling the model to more accurately capture students' knowledge mastery and learning needs.

Therefore, the KRMKG-GNN model introduced in this study represents a significant advancement in the field of educational technology, particularly for Network Engineering education. By integrating text, image, and behavioral data into a comprehensive Multimodal Knowledge Graph, the model has demonstrated exceptional performance in personalized learning content recommendation, achieving a remarkable 92.7% accuracy rate. This research not only outperforms existing models like the Graph Attention Network (GAT) in terms of knowledge point coverage and error rate reduction but also addresses the critical need for dynamic personalization in learning experiences. The implications of this work are profound, offering a foundation for future research to expand the model's capabilities, including incorporating more diverse data types and

assessing its scalability across various educational contexts. The KRMKG-GNN model is a testament to the potential of intelligent systems in enhancing teaching effectiveness and student learning outcomes, and it is poised to play a pivotal role in shaping the future of personalized education.

Author contributions

LZ-Writing-original draft, review and editing, Conceptualization, Formal analysis, Methodology, Validation, Writing-review and editing, Supervision, Visualization, Project administration.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study did not involve any human or animal experimentation.

Competing interests

The authors declare no competing interests.

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